Updated on 21 June 2026

Curriculum vitae

김승원(金勝援)

기계항공공학박사

Seung-Won Kim, Ph.D.



Google Scholar

Senior Research Scientist (선임연구원)

Intelligence and Interaction Research Center (지능·인터랙션연구센터)

Advanced Materials and Systems Research Division (첨단소재·시스템연구본부)

Korea Institute of Science and Technology (KIST) (한국과학기술연구원)

& Associate Professor (부교수)

Division of AI-Robotics

University of Science & Technology (UST) & KIST School

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Address: Bld.# L8, Rm.# 505, International Cooperation Center, Korea Institute of Science and Technology (KIST), Hwarang-ro 14-gil 5, Seongbuk-gu, Seoul, 02792, Republic of Korea

EDUCATION

Seoul National University

B.S., School of Mechanical & Aerospace Engineering *1, Mar. 2005 ~ 26, Feb. 2009*

Ph.D., School of Mechanical & Aerospace Engineering *1, Mar. 2009 ~ 26, Feb. 2016*

Advisor: Prof. Kyu-Jin Cho

Dissertation: Bend Propagating Actuation Utilizing a Characteristic of Developable Surface of Bistable Structure

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RESEARCH CAREER

Intelligence and Interaction Research Center in Korea Institute of Science and Technology, Seoul, Korea

Senior Researcher *Sept. 2024 – Present*

- Working for developing soft robots, smart actuators and sensors, and healthcare service robots

Center for Healthcare Robotics in Korea Institute of Science and Technology, Seoul, Korea

Senior Researcher *Mar. 2018 – Aug. 2024*

- Working for developing medical & healthcare service robots
- Working for Human Augmentation Technology

Soft Machines Lab at Department of Mechanical Engineering, Carnegie Mellon University, PA, U.S.A.

Visiting Scholar working with Prof. Carmel Majidi *Jan. 2020 – Dec. 2020*

- Worked for developing variable stiffness mechanism utilizing low melting point material

Healthcare Robotics Research Group in Korea Institute of Science and Technology, Seoul, Korea

Researcher *June. 2016 – Feb. 2018*

- Worked for developing healthcare service robots

Institute of Advanced Machinery and Design in Seoul National University, Seoul, Korea

Senior Researcher under Prof. Kyu-Jin Cho *Mar. 2016 – Apr. 2016*

- Worked for developing soft manipulator and robots
- Investigated and assisted for planning a proposal of Engineering Research Center for soft robotics organized by National Research Foundation of Korea

Biorobotics Laboratory in Seoul National University, Seoul, Korea

Research Assistant under Prof. Kyu-Jin Cho *Mar. 2009 – Feb. 2016*

- Worked for developing soft bio-inspired robots and mechanisms
- Assisted for planning research proposals and writing research reports

SCHOLARSHIP

Global Ph.D. Fellowship

- Scholarship funded by National Research Foundation of Korea (Mar. 2011 ~ Feb. 2014)
- Financial support to Ph.D. candidates for outstanding research proposal & achievements
- Research Title “Development of Core Elemental Technology for Multi-scale Bio-inspired Robot”

Brain Korea 21

- Scholarship funded by National Research Foundation of Korea (Sept. – Dec. 2010)
- Financial support to graduate students with outstanding academic achievements

Support of Lecture & Research

- Scholarship funded by Seoul National University (Sept. – Dec. 2009)
- Financial support to graduate students for lecture & research assistance

Superior Academic Performance

- Scholarship funded by Seoul National University
- Financial support to students with outstanding academic achievements
- One year of full funding(Sept. 2005 – June 2006) and 2 years of partial funding(Sept. – Dec. 2006, Sept. 2007 – Dec. 2008)

HONOR & AWARDS

1. **Research Development Team Award** about "MIDAS Project," Korea Institute of Science and Technology (KIST), Feb. 2023.
2. **Research Development Team Award** about "Surgery4.0 Project," Korea Institute of Science and Technology (KIST), Feb. 2020.
3. **Unsung Hero** about Living Lab., Korea Institute of Science and Technology (KIST), Dec. 2019.
4. **Research Development Team Award** about "Human Plus R&D Project", Korea Institute of Science and Technology (KIST), Feb. 2018.
5. **Outstanding Presenter Award**, International Symposium on Green Manufacturing and Applications (ISGMA), 2015.
6. **Model cases of Creative Synergy Converging Research to open a new future**, Project "Fundamental Technology for Biomimetic Soft Morphing" accomplishment was awarded by Convergence Research Policy Center in Korea Institute of Science and Technology (KIST), 2014.
7. **Best Paper Award**, Autumn conference of the Korean Society for Composite Materials, 2013.
8. **Best Poster Award**, International Symposium on Green Manufacturing and Applications (ISGMA), 2012.
9. **Best Paper Award in Robot & Automation**, Spring Conference of the Korean Society for Precision Engineering, 2010.
10. **Best Student Paper Award**, IEEE/RAS-EMBS International Conference on Biomedical Robotics and Biomechatronics (BIOROB), 2010.

RESEARCH PROJECT ACTIVITIES

Development and integration of elements of robot technology for tomato harvest in smartfarm

June 2025 – Dec. 2026

- Supported by Korea Institute of Science and Technology(KIST)
- Project director

Development of a variable stiffness endoscope mechanism utilizing thermoelectric devices and phase transition materials

Mar. 2021 – Feb. 2026

- Supported by the National Research Foundation of Korea(NRF)

grant funded by the Korea government(Ministry of Science and ICT)

- Project director

Development of Fundamental Technology of Medical Intelligence for Digitally-Assisted Surgery(MIDAS) to lead the era of Surgery 4.0 Jan. 2021 – Dec. 2025

- funded by Korea Institute of Science and Technology(KIST)
- participating researcher

CAS-surgery4.0: target-selective fluorescence imaging guided micro surgical robots Jan. 2018 – Dec. 2020

- funded by Korea Institute of Science and Technology(KIST)
- participating researcher

KIST Joint Research Lab. July 2018 – Dec. 2020

: Fundamental Research on Soft Bio-inspired Robotics Technology for Medical Robots

- funded by Korea Institute of Science and Technology(KIST)
- participating as a cooperating researcher

Development of Fundamental Core Technology of Smart Garment for Physical Optimization April 2018 – Nov. 2018

- funded by National Research Foundation of Korea, Ministry of Science and ICT
- participating researcher

Project Planning for Development of Convergence Technology of Human Augmentation April 2017 – Dec. 2017

- funded by National Research Foundation of Korea, Ministry of Science and ICT
- participating researcher

Development of Robot Systems for Total Nursing Service June 2016 – May. 2018

- funded by Korea Evaluation Institute of Industrial Technology, Ministry of Trade, Industry and Energy
- participating researcher

Development of Fundamental Technology of Soft Robotics for Advanced Soft Gripper June 2015 – Apr. 2016

- funded by National Research Foundation of Korea, Ministry of

Science, ICT and Future Planning
- participating researcher

Fundamental Technology for Active Morphing Flexible Display Oct. 2014 – Oct. 2015

- funded by LG Display
- participating researcher

Biomimetic Robot Research Center Oct. 2013 – Dec. 2015

- funded by Defense Acquisition Program Administration & Agency for Defense Development
- participating researcher

Development of Core Elemental Technology for Multi-scale Bio-inspired Robot Mar. 2011 – Feb. 2013

- Global Ph.D. Fellowship funded by National Research Foundation of Korea, Ministry of Science, ICT and Future Planning
- project director

Development of Multi-functional Endoskeleton for Biomimetic Robots May 2009 – April 2011

- funded by National Research Foundation of Korea, Ministry of Science, ICT and Future Planning
- participating researcher

Fundamental Technology for Biomimetic Soft Morphing Sept. 2009 – June 2014

- funded by National Research Foundation of Korea, Ministry of Science, ICT and Future Planning
- participating researcher
- managing project as a student representative for 3 years (July 2011 – June 2014)

Developing of Design and Manufacturing for Multi-scale Mass-deployable cooperative robots Sept. 2009 – Mar. 2013

- funded by National Research Foundation of Korea, Ministry of Science, ICT and Future Planning
- participating researcher

TEACHING EXPERIENCE

UST-KIST School, Korea

Teaching, “Introduction to Nano & Information Technology” May – June, 2021
May – June, 2019

- Lecture for introduction of robotics

Teaching, "Tech-Bridge: AI-Robotics" 1st semester, 2025

- Invited seminar styled lecture for introduction of AI-Robotics

Kyung Hee University Global Campus, Korea April 7, 2016
Mar. 31, 2015

Seminar on "Bio-inspired robots and Artificial Muscle Actuators"

- One-day lecture for 30 undergraduate students

Seoul National University, Korea Sept. – Dec. 2012

Teaching Assistant, "Biologically Inspired Robotics" (Prof. Kyu-Jin Cho)

- Prepared of the lecture note and graded the homework
- Opening TA session for 13 graduate students

Seoul National University, Korea Sept. – Dec. 2009

Teaching Assistant, "Dynamics" (Prof. Kyu-Jin Cho)

- Prepared of the lecture note and graded the homework
- Opening TA session for 90 undergraduate students

Seoul National University, Korea Mar. – June 2009

Teaching Assistant, "System Analysis in Mechanical and Aerospace Engineering" (Prof. Kyu-Jin Cho)

- Prepared of the lecture note and graded the homework
- Opening TA session for 106 undergraduate students

Seoul National University, Korea Mar. 2019 – Feb. 2020
Mar. 2012 – Feb. 2013
Mar. 2009 – Feb. 2011

B.S. Thesis Tutor

- Led the B.S. Thesis of 7 undergraduate students
- 5 students received the outstanding B.S. Thesis award

ACADEMIC SERVICES

Member

- IEEE Member, *Sept. 2010 – Present*
  (Robotics and Automation Society, Bio Robotics TC, Soft Robotics TC, Wearable Robotics TC)
- Korea Robotics Society, Member, *Jan. 2017 – Present*
- The Korean Society of Mechanical Engineers (Dynamics, control, robot), Member, *2023 – Present*
- Korean CDE Society, Life member, *Jan. 2025 – Present*

Reviewer of Academic Journals and Conferences

- IEEE Transactions on Robotics (TRO)
- IEEE/ASME Transactions on Mechatronics (TMECH)
- IEEE Robotics and Automation Letter (RA-L)
- IEEE Transactions on Neural Systems & Rehabilitation Engineering (TNSRE)
- Smart Materials and Structures (SMS)
- Bioinspiration & Biomimetics (BIOBIO)
- Soft Robotics (SORO)
- Textile Research Journal
- ASME Journal of Dynamic Systems, Measurement, and Control
- Journal of Mechanical Science and Technology (JMST)
- Journal of Computational Design and Engineering (JCDE)
- International Journal of Precision Engineering and Manufacturing (IJPEM)
- Intelligent Service Robotics (ISR)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Conference on Soft Robotics (RoboSoft)
- IEEE International Conference on Ubiquitous Robots (UR)

TECHNICAL SKILLS

● Manufacturing & Experiment technologies

A. *Smart Composite Microstructure(SCM) process* with various composite materials including glass fiber and carbon fiber reinforced prepreg(CFRP)

B. *Shape Deposition Manufacturing(SDM)* with rapid prototyping(*RP*) & Polymer molding

C. *Shape Memory Alloy(SMA) actuator design and heat annealing process*

D. *Fabrication devices* (CNC milling/drilling/lathe, 3D printing, heat pressing, laser cutting, heat annealing with oven or furnace, auto-clave, spin-coating)

E. *Experiment devices* (Universal tensile test machine, High speed camera, Differential scanning calorimetry(DSC))

● Software

A. *Computer aided design, CAD* (Solidworks, PowerMill, Visual CAM, Eagle Layout Editor)

B. *Numerical analysis* (MATLAB, COMSOL MultiPhysics)

C. *Data analysis* (Origin, SPSS)

D. *Embedded control & DAQ* (NI LabView & CompactRIO, Flexlogger, Delsys Trigno)

E. *Video & Image processing* (Adobe Premiere Pro & Photoshop, ProAnalyst)

PUBLICATIONS

● International Journals

2025

1. Jae-Kyeong Kim, Se Hyeok Ahn, Sun-Pill Jung, Je-Moon Kim, Deuk-Gyeom Hwang, Ha-Seon Kim, **Seung-Won Kim**, Dae-Young Lee, Jinkyu Yang, and Kyu-Jin Cho, "Double-layer self-locking origami based on opposite folding motion," *Soft Robotics (SORO)*, 21695172251401337, Dec. 2025.
2. Hyungseok Yoon and **Seung-Won Kim**, "Efficient and robust fabrication of soft sensors via injection with auxiliary suction in multilayered microchannels with a liquid metal alloy," *IEEE Sensors Journal (JSEN)*, **25**(13), pp. 23948-23957, July 2025. [SCIE]
3. Jae-Ryeong Choi, **Seung-Won Kim***, and Kyu-Jin Cho*, "Self-unlocking active clutch for quasi-passive wearable robots," *IEEE/ASME Transactions on Mechatronics (TMECH)*, **30**(3), pp. 2401-2410, June 2025. [SCIE] (*Co-corresponding authorship)

2024

4. Hang Man Cho, Jae-Ryeong Choi, Jung-Hwan Moon, Kyu-Jin Cho, and **Seung-Won Kim**, "Evaluation of an assistive exosuit for alleviating neck and shoulder muscle fatigue during prolonged flexed neck posture," *Journal of NeuroEngineering and Rehabilitation (JNER)*, **21**, 232, Dec. 2024. [SCIE]
5. Chaegyu Lim, Me-Yeon Lee, and **Seung-Won Kim**, "Recent development of medical service robots in Republic of Korea and their field demonstration cases in the clinical setting," *JMST Advances*, **6**, pp. 395-401, Nov. 2024.
6. Myungjoong Lee, **Seung-Won Kim**, and Hyunki In, "Archi-TangiBlock: a modular block based tangible media as a CAD tool for visually impaired people," *IEEE Access*, **12**, pp. 46582-46595, Mar. 2024. [SCIE]
7. Youngsu Cha, Kwang Jin Kim**, Seung-Won Kim**, and Hyosang Lee, "Soft component technology and application for soft robotics," *Intelligent Service Robotics (ISR)*, **17**(1), pp. 1-2, Jan. 2024. [SCIE] (Editorial Material))

2021

8. Choong Sun Kim, Ock Kyun Oh, Hyeongdo Choi, Yong Jun Kim, Gyu Soup Lee, Hyun Jeong Kim, Carmel Majidi, **Seung-Won Kim***, and Byung Jin Cho*, "Variable rigidity module with a flexible thermoelectric device for bi-directional temperature control," *Soft Robotics (SORO),* **8**(6), pp. 662-672, Dec. 2021. [SCI] (*Co-corresponding authorship)
9. Hyeonjung Lim, **Seung-Won Kim**, Jae-Bok Song, and Youngsu Cha, "Thin piezoelectric mobile robot using curved tail oscillation," *IEEE Access,* **9**, pp. 145477-145485, Oct. 2021. [SCIE]

2020

10. **Seung-Won Kim***, Sung Hee Kim*, Choong Sun Kim, Kyoungsoo Yi, Jun-Sik Kim, Byung Jin Cho, and Youngsu Cha, "Thermal display glove for interacting with virtual reality," *Scientific Reports*, **10**, 11403, July 2020. [SCIE] (Co-first authorship)

2015

11. **Seung-Won Kim**, Jong-Gu Lee, Sungmin An, Maenghyo Cho, and Kyu-Jin Cho, "A large-stroke shape memory alloy spring actuator using double-coil configuration," *Smart Materials and Structures (SMS),* **24**(9), 095014, Aug. 2015. [SCI]
12. Jong-Gu Lee, Junghyun Ryu, **Seung-Won Kim**, Je-Sung Koh, Kyu-Jin Cho, and Maenghyo Cho, "Effect of initial tool-plate curvature on snap-through load of unsymmetric laminated cross-ply bistable

composites," *Composite Structures (CS),* **122**, pp. 82-91, Apr. 2015. [SCIE]

2014

13. Junghyun Ryu, Jong-Gu Lee, **Seung-Won Kim**, Je-Sung Koh, Kyu-Jin Cho, and Maenghyo Cho, "Generalized curvature tailoring of bistable CFRP laminates by curing on a cylindrical tool-plate with misalignment," *Composites Science and Technology (CST),* **103**, pp. 127-133, Aug. 2014. [SCI]
14. **Seung-Won Kim**, Je-Sung Koh, Jong-Gu Lee, Junghyun Ryu, Maenghyo Cho, and Kyu-Jin Cho, "Flytrap-inspired robot using structurally integrated actuation based on bistability and a developable surface," *Bioinspiration & Biomimetics (BIOBIO),* **9**(3), 036004, Mar. 2014. [SCIE]

2013

15. Junghyun Ryu, Jung-Pyo Kong, **Seung-Won Kim**, Je-Sung Koh, Kyu-Jin Cho, and Maenghyo Cho, "Curvature tailoring of unsymmetric laminates with an initial curvature," *Journal of Composite Materials (JCM),* **47**(25), pp. 3163-3174, Nov. 2013. [SCI]
16. Dae-Young Lee, Je-Sung Koh, Ji-Suk Kim, **Seung-Won Kim**, and Kyu-Jin Cho, "Deformable-wheel Robot based on Soft Material," *International Journal of Precision Engineering and Manufacturing (IJPEM),* **14**(8), pp. 1439-1445, Aug. 2013. [SCIE]

2012

17. Minkyun Noh, **Seung-Won Kim**, Sungmin An, Je-Sung Koh, and Kyu-Jin Cho, "Flea-Inspired catapult mechanism for miniature jumping robots," *IEEE Transactions on Robotics (TRO),* **28**(5), pp. 1007-1018, Oct. 2012. [SCI]

● International Conferences

2025

1. Hyungseok Yoon and **Seung-Won Kim**, "Efficient and robust fabrication of soft sensors via injection with auxiliary suction in multilayered microchannels with a liquid metal alloy," *IEEE Sensors 2025 Conference*, Oct. 2025. (Pre-published paper presentation)

2019

2. **Seung-Won Kim**, "Thermomechanical characteristics of a shape memory polymer for a variable stiffness mechanism in medical applications," *The 4th International Conference on Active Materials and Soft Mechatronics (AMSM)*, Oct. 2019.
3. **Seung-Won Kim,** "Modular and affordable power assist module for various physical care services," *The 13th International Society of Physical and Rehabilitation Medicine World Congress (ISPRM)*, June 2019.
4. **Seung-Won Kim**, Sung Hee Kim, and Youngsu Cha, "A thermal stimulating glove for virtual reality," *2019 The 2nd IEEE International Conference on Soft Robotics (RoboSoft)*, *Late Breaking Results*, Apr. 2019.

2018

5. **Seung-Won Kim**, "Variable stiffness mechanism with low melting point materials for medical applications," *The 3rd International Workshop on Active Materials and Soft Mechatronics (AMSM)*, Oct. 2018.
6. **Seung-Won Kim**, Jiyeon Song, Seungbeum Suh, Woosub Lee, and Sungchul Kang, "Design and experiment of a passive sit-to-stand and walking (STSW) assistance device for the elderly," *40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*,

July 2018.

2017

7. Jiyeon Song, **Seung-Won Kim**, Dong-Eun Choi, Gang-Tae Bae, Woosub Lee, and Sung Chul Kang, "Development of a passive standing-up assistance device for the elderly," *The 2017 IEEE International Conference on Multisensor Fusion and Integration for Intelligent systems (MFI)*, Nov. 2017.
8. Gang-Tae Bae, **Seung-Won Kim**, Dong-Eun Choi, Changhyun Cho, Woosub Lee, and Sung Chul Kang, "Omni-directional power-assist-modular(PAM) mobile robot for total nursing service system," *14th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI)*, June 2017.

2015

9. **Seung-Won Kim** and Kyu-Jin Cho, "A bistable mechanism using residual stress for morphing structures," *2015 International Symposium on Green Manufacturing and Applications (ISGMA)*, June 2015. **Outstanding Presentation Award**

2014

10. Jong-Gu Lee, Junghyun Ryu, Maenghyo Cho, **Seung-Won Kim**, and Kyu-Jin Cho, "Evaluation of initial curvature effect on the snap-through load of bi-stable composites," *55th AIAA/ASME/ASCE/ AHS/ASC Structures, Structural Dynamics, and Materials Conference (AIAA SciTech)*, Jan. 2014.

2013

11. **Seung-Won Kim**, Jun-Young Lee, and Kyu-Jin Cho, "Towards a bistable morphing winglet for unmanned aerial vehicle (UAV)," *2013 The 44th International Symposium on Robotics (ISR)*, Oct. 2013.
12. Jong-Gu Lee, Junghyun Ryu, **Seung-Won Kim**, Kyu-Jin Cho, and Maenghyo Cho, "A study on the design guideline of SMA spring with the initial curvature added bi-stable composite," *2013 International Conference on Advances in Interaction & Multiscale Mechanics (AIMM)*, Sept. 2013.
13. Jung-Young Lee, **Seung-Won Kim**, and Kyu-Jin Cho, "Venus flytrap inspired bistable morphing winglet," *2013 The 7th World Congress on Biomimetics, Artificial Muscles and Nano-Bio (BAMN)*, Aug. 2013.
14. Junghyun Ryu, Jong-Gu Lee, **Seung-Won Kim**, Jun-Young Lee, Kyu-Jin Cho, and Maenghyo Cho, "Unexpected twisting curvature generation of bistable CFRP laminate due to the uncertainty of lay-up sequence and negative initial curvature," *The 19th International Conference on composite Materials (ICCM)*, July 2013.
15. Je-Sung Koh, Sunpill Jung, Minkyun Noh, **Seung-Won Kim**, and Kyu-Jin Cho, "Flea inspired catapult mechanism with active energy storage and release for small scale jumping robot," *Proceedings of the 2013 IEEE International Conference on Robotics and Automation (ICRA)*, May 2013.

2012

16. Jong-Gu Lee, Junghyun Ryu, **Seung-Won Kim**, Kyu-Jin Cho, and Maenghyo Cho, "Analysis of the critical moment triggering off snap-through of bistable composite with initial curvature," *2012 2nd International Conference on Computational Design in Engineering (CDE)*, Nov. 2012.
17. Junghyun Ryu, Jong-Gu Lee, **Seung-Won Kim**, Je-Sung Koh, Kyu-Jin Cho, and Maenghyo Cho, "Instability of principal curvature direction of unsymmetric cross-ply laminates due to uncertainty of lay-up sequence," *2012 2nd International Conference on Computational Design in Engineering (CDE)*, Nov. 2012.
18. Je-Sung Koh, Dae-Young Lee, **Seung-Won Kim**, and Kyu-Jin Cho, "Deformable soft wheel robot using hybrid actuation," *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct. 2012.

2011

19. Je-Sung Koh, **Seung-Won Kim**, Minkyun Noh, and Kyu-Jin Cho, "Biologically inspired robots using smart composite microstructures," *2011 8th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI)*, Nov. 2011.
20. **Seung-Won Kim**, Je-Sung Koh, and Kyu-Jin Cho, "Active morphing robot inspired by the pre-strained fiber structure of the Venus flytrap," *2011 8th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI)*, Nov. 2011.
21. **Seung-Won Kim**, Je-Sung Koh, and Kyu-Jin Cho, "Design & Analysis a flytrap robot using bi-stable composite," *Proceedings of the 2011 IEEE International Conference on Robotics and Automation (ICRA)*, May 2011.
22. Minkyun Noh, **Seung-Won Kim**, and Kyu-Jin Cho, "A miniature jumping robot with flea-inspired catapult system: active latch and trigger," *Proceedings of 2011 International Workshop on Bio-inspired Robots*, Apr. 2011.

2010

23. **Seung-Won Kim**, Je-Sung Koh, Maenghyo Cho, and Kyu-Jin Cho, "Towards a bio-mimetic flytrap robot based on a snap-through mechanism," *Proceedings of the 2010 3rd IEEE/RAS-EMBS International Conference on Biomedical Robotics and Biomechatronics (BIOROB)*, Sept. 2010. **Best Student Paper Award**

● Domestic Journals

1. Chaegyu Lim, Me Yeon Lee, and **Seung-Won Kim**, "Recent development of medical service robots in Republic of Korea and their field demonstration cases in the clinical setting,", *Journal of the KSME*, Dec. 2024. (Note: Korea translational edition of the JMST Advances article)
2. Jun-Sik Kim, Dong Hwan Kim, **Seung-Won Kim**, Deukhee Lee, and Jongsuk Choi, "Robot research in Korea Institute of Science and Technology: Present and future," *Korea Robotics Society Review*, **17**(4), pp. 3-11, Oct. 2020.
3. Je-Sung Koh, Dae-Young Lee, Ji-Suk Kim, **Seung-Won Kim**, and Kyu-Jin Cho, "Design and fabrication of soft deformable wheel robot using composite materials and shape memory alloy coil spring actuators," *Journal of the Korean Society for Precision Engineering*, **30**(1), pp. 47-52, Jan. 2013.
4. **Seung-Won Kim**, Je-Sung Koh, Maenghyo Cho, and Kyu-Jin Cho, "Soft morphing motion of flytrap robot using bending propagating actuation," *Journal of Institute of Control, Robotics and Systems*, **18**(3), pp. 168-174, Mar. 2012.
5. Je-Sung Koh, **Seung-Won Kim**, and Kyu-Jin Cho, "Biomimetic miniature jumping robot," *Robot and Human*, **9**(1), pp. 3-12, Feb. 2012.

● Domestic Conferences

2026

1. Geun Taek Bae and **Seung-Won Kim**, "Design of a scoop-type pedicel cutting mechanism for tomato harvesting robots," *The 21th Korea Robotics Society Annual Conference (KROC)*, Feb. 2026.

2025

2. Geun Taek Bae*, Seong Hun Jo*, and **Seung-Won Kim**, “Design of a distal pedicel cutting mechanism for automated tomato harvesting robots,” *80th Anniversary Conference of Korea Society of Mechanical Engineering (KSME)*, Dec. 2025. (*Co-first authorship)
3. Chaegyu Lim, Yongsu Cha, and **Seung-Won Kim**, “Level change of shear thickening fluid due to vibration stimulation,” *Spring Conference in Dynamics, Controls, and Robotics field of Korea Society of Mechanical Engineering (KSME)*, Apr. 2025.
4. **Seung-Won Kim**, “Development and performance evaluation of a wearable device for reducing muscle fatigue in the neck and shoulder regions during static posture maintenance,” *2025 Winter Conference of Society for Computational Design and Engineering (CDE)*, Feb. 2025.

2024

5. **Seung-Won Kim**, Hang Man Cho, and Junghwan Moon, “Design and usability evaluation of a wearable device for muscle fatigue reduction considering body characteristics and work environment,” *2024 Summer Conference of Society for Computational Design and Engineering (CDE)*, Aug. 2024.
6. Hyungseok Yoon and **Seung-Won Kim**, “Effect of injection with assistive suction in the fabrication of liquid metal based soft sensors,” *Spring Conference in Dynamics, Controls, and Robotics field of Korea Society of Mechanical Engineering (KSME)*, Apr. 2024.
7. **Seung-Won Kim** and Hang Man Cho, “Analysis of required supporting force and muscle fatigue reduction by supporting neck flexion,” *Spring Conference in Dynamics, Controls, and Robotics field of Korea Society of Mechanical Engineering (KSME)*, Apr. 2024.
8. Jae-Kyeong Kim, Sun-Pill Jung, Se Hyeok Ahn, Deuk-Gyeom Hwang, **Seung-Won Kim**, Dae-Young Lee, Jinkyu Yang, and Kyu-Jin Cho, “Design of a lightweight origami arm support device for overhead tasks,” *The 19th Korea Robotics Society Annual Conference (KROC)*, Feb. 2024.
9. Junghwan Moon and **Seung-Won Kim**, “Design of a head immobilizer using a dual-layered particle jamming mechanism for facial treatment,” *The 19th Korea Robotics Society Annual Conference (KROC)*, Feb. 2024.

2023

10. Hang Man Cho and **Seung-Won Kim**, “Structural analysis of variable stiffness fixture with thermoplastic for wearable robots,” *2023 Summer Conference of Society for Computational Design and Engineering (CDE)*, Aug. 2023.
11. Hyungseok Yoon and **Seung-Won Kim**, “Evaluation of structural durability and reliability with respect to the manufacturing method of a soft sensor based on liquid metal,” *The 18th Korea Robotics Society Annual Conference (KROC)*, Feb. 2023.
12. Junghwan Moon and **Seung-Won Kim**, “A satisfaction survey on the fixed load of a head fixation device for skin laser treatment,” *The 18th Korea Robotics Society Annual Conference (KROC)*, Feb. 2023.
13. Hang Man Cho, Jae-Ryeong Choi, Su Youn Park, Kyu-Jin Cho, and **Seung-Won Kim**, “Development of a wire-driven neck assistive device for supporting neck muscles of surgeons,” *The 18th Korea Robotics Society Annual Conference (KROC)*, Feb. 2023.

2022

14. Hyungseok Yoon and **Seung-Won Kim**, “Comparison of structural characteristics depending on fabrication methods of a soft sensor based on liquid metal (EGaIn),” *2022 Autumn Conference of the Korean Society for Precision Engineering (KSPE)*, Oct. 2022.
15. Junghwan Moon and **Seung-Won Kim**, “Design and analysis of a head fixation device for laser skin treatment,” *2022 Autumn Conference of the Korean Society for Precision Engineering(KSPE)*, Oct. 2022.

16. Hangman Cho, Jae-Ryeong Choi, Kyu-Jin Cho, and **Seung-Won Kim**, "Design of a wire-driven neck supporting wearable device for surgeons using ratchet-gear-based locking mechanism," *2022 Autumn Conference of the Korean Society for Precision Engineering (KSPE)*, Oct. 2022.

2021

17. **Seung-Won Kim**, "Thermomechanical analysis of a variable stiffness mechanism using thermoplastic and thermoelectric module for endoscopic applications," *Computational Structural Engineering Institute of Korea (COSEIK) Academic Symposium*, Nov. 2021.
18. **Seung-Won Kim**, Myoungju Kim, Juyeong Kim, and Jaehyeon Park, "Heat transfer analysis of a variable stiffness endoscope utilizing thermoplastic and thermoelectric device inside a large intestine," *The 16th Korea Robotics Society Annual Conference (KROC)*, May 2021.

2019

19. **Seung-Won Kim*** and Ki Hun Kang, "Analysis of tensile·thermal properties of Field's metal for design of variable stiffness mechanism," *The 14th Korea Robotics Society Annual Conference (KROC)*, Jan. 2019.

2018

20. Ganghyun Jeon, **Seung-Won Kim**, and Yong-Jai Park, "Study on the fabrication of bistable structure using 3D printer," *2018 Spring Conference of the Korean Society for Precision Engineering (KSPE)*, May 2018.
21. **Seung-Won Kim**, Jiyeon Song, Seungbeum Suh, Woosub Lee, and Sung-Chul Kang, "A passive sit-to-stand & walk assistive device for indoor movements of the elderly," *The 13th Korea Robotics Society Annual Conference (KROC)*, Jan. 2018.

2017

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